import streamlit as st import math import heapq import networkx as nx import matplotlib.pyplot as plt # import the necessary functions and variables from searchAlgos.py from searchAlgos import (hospital_graph, locations, gbfs, a_star) # Streamlit GUI #********************# # Set Page Config # write meaningful title and description for the app # define the nodes and their coordinates nodes = list(hospital_graph.keys()) # create a selectbox for the user to choose the start and goal nodes start = st.selectbox("Select Initial Node", #pass the list of nodes to the selectbox # set the default value to "Pharmacy") goal = st.selectbox("Select Goal Node", #pass the list of nodes to the selectbox # set the default value to "Emergency_Ward") # create a selectbox for the user to choose the search algorithm if st.button("Run Search"): if algorithm == "GBFS": # run the GBFS algorithm with the selected start and goal nodes pass else: # run the A* algorithm with the selected start and goal nodes pass if path is None: # display a error message indicating that no path was found pass else: # Display result st.subheader("Search Result") st.write(f"**Algorithm:** {algorithm}") st.write(f"**Solution Path:** {' → '.join(path)}") st.write(f"**Total Path Cost:** {cost:.2f}") # Visualize NetworkX graph G = nx.DiGraph() for node, neighbors in hospital_graph.items(): for neighbor, weight in neighbors.items(): pass pos = locations fig, ax = plt.subplots(figsize=(10, 6)) ax.set_title(f"{algorithm} Solution Path") ax.axis("off") st.pyplot(fig)